CS383 · SOFTWARE ENGINEERING

Midterm Model 3 Practice

Undated · 20 marks · four parts A to D

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01 About this midterm

A midterm out of 20, split into four parts. The photos show no date and no term number, so I named it Model 3. Its format differs from the others, so it most likely belongs to another instructor or an older term. The questions are copied from photos of a student's solved paper.

Part	A	B	C	D	Total
Marks	9	3	4	4	**20**

About the solutions

The published paper carries the note "the solutions are not 100% correct", and it has no instructor's grading. So I did not rely on the student's answers. I solved it myself from the slides, and I point out every item where my answer differs from the student's.

02 Questions

A MCQ · 9 M

Part A) In the following multiple-choice items, put a circle around the letter of the correct choice. There is only one correct answer per item.

A1) The following is an example of a generic type of software product:

a. patient registration application of a private hospital

b. Microsoft PowerPoint

c. a stock prediction application for investment in a company

A2) Motivation behind software engineering includes:

a. to rapidly produce large software system

b. to handle development and maintenance cost

c. to distinguish market needs of software

A3) The following is NOT part of the software engineering process activity:

a. software design b. software specification c. software management

A4) In the Scrum management framework, the inputs of sprint planning include:

a. team capacity b. sprint goal c. sprint backlog

A5) Product quality relies on process quality, management quality, and:

a. development technology b. fulfilment of clients' requirements c. short delivery window

A6) One of the essential software design activities is:

a. requirements design b. architectural design c. test design

B TRUE / FALSE WITH REASON · 3 M

Part B) Determine whether the following statements are valid or not. Select true or false and then support your answer with a reason regardless of the state.

B1) It is considered that software development has more cost (timewise) than software maintenance.

True | False Reason:

B2) The initial estimation of engineers' efforts toward software development is always accurate.

True | False Reason:

C PROCESS MODEL · 4 M

Part C) Complete the following table and select which software process model is appropriate for the given concern. You must provide a reason for your selection.

C1) The target system is intended for safety-critical operations.

Choose one: ○ Waterfall ○ Agile ○ Component-based Reason:

C2) The clients need to see prototypes of the developed system and provide early feedback to the software engineers.

Choose one: ○ Waterfall ○ Agile ○ Component-based Reason:

D ETHICS · 4 M

Part D) Answer the following questions that are related to Software Engineering Ethics and Professional Practices (SEEPP):

D1) [2 marks] One principle that SEEPP covers is Public, which states that software engineers shall act consistently in the public interest. Give two examples of adhering to this principle.

D2) [2 marks] In the following table, indicate which principle is best submitted to the provided example. Principles that can be considered include:

1. Management: promote an ethical approach to managing software development.
2. Profession: advance the profession's integrity consistent with the public interest.
3. Judgment: maintain integrity, morals, and independence in their professional judgment.
4. Self: participate in lifelong learning of the practice of their profession.

Example	Principle
D2.1) A software engineer must ensure a realistic estimation of cost and plan on all projects.	
D2.2) A software engineer shouldn't illegally obtain a software application for any purpose.	